Fourier Analysis is a mathematical tool used to decompose functions into sums or integrals of simpler trigonometric functions. It is fundamental in physics for analyzing waveforms, signals, and various physical phenomena. This section will cover the Fourier Series for periodic functions and the Fourier Transform for non-periodic functions, with examples from physics to illustrate these concepts.

0.1 Fourier Series

A Fourier Series allows us to represent a periodic function as a sum of sine and cosine functions. This is particularly useful in physics, where many systems exhibit periodic behavior (e.g., vibrations, waves, and alternating currents).

Concept of Fourier Series

For a periodic function $f(x)$ with period T, the Fourier Series representation is given by:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{2\pi n x}{T}\right) + b_n \sin\left(\frac{2\pi n x}{T}\right) \right),$$

where the coefficients a_n and b_n are calculated as:

$$a_0 = \frac{1}{T} \int_0^T f(x)\, dx,$$

$$a_n = \frac{2}{T} \int_0^T f(x) \cos\left(\frac{2\pi n x}{T}\right) dx, \quad n \geq 1,$$

$$b_n = \frac{2}{T} \int_0^T f(x) \sin\left(\frac{2\pi n x}{T}\right) dx, \quad n \geq 1.$$

Convergence of Fourier Series

The convergence of a Fourier Series depends on the properties of the function $f(x)$. For functions that are piecewise continuous and have a finite number of discontinuities, the Fourier Series converges to $f(x)$ at all points where $f(x)$ is continuous. At points of discontinuity, the series converges to the average of the left-hand and right-hand limits (known as the Dirichlet conditions).

Physics Insight: In physical systems, such as vibrating strings or sound waves, the functions describing these phenomena are often piecewise continuous. Therefore, their Fourier Series will converge to the actual physical displacement or pressure at most points.

Orthogonality of Sine and Cosine Functions

The sine and cosine functions used in Fourier Series are orthogonal over a period T:

$$\int_0^T \cos\left(\frac{2\pi n x}{T}\right) \cos\left(\frac{2\pi m x}{T}\right) dx = 0 \quad \text{if } n \neq m,$$

$$\int_0^T \sin\left(\frac{2\pi n x}{T}\right) \sin\left(\frac{2\pi m x}{T}\right) dx = 0 \quad \text{if } n \neq m.$$

This orthogonality is key to calculating the Fourier coefficients because it ensures that each sine or cosine term in the series represents a distinct frequency component of the function.

Physics Example: Vibrating String Consider a vibrating string of length L fixed at both ends. The displacement $y(x,t)$ of the string at position x and time t can be described as a superposition of standing wave modes:

$$y(x,t) = \sum_{n=1}^{\infty} \left(A_n \cos\left(\frac{n\pi x}{L}\right) \cos(\omega_n t) + B_n \sin\left(\frac{n\pi x}{L}\right) \sin(\omega_n t) \right).$$

Here, $\omega_n = \frac{n\pi v}{L}$ is the angular frequency of the n-th mode, and v is the wave speed. The coefficients A_n and B_n represent the amplitudes of the cosine and sine components, respectively.

Key Insight: The Fourier Series allows us to decompose the complex motion of a vibrating string into a sum of simpler sinusoidal motions. Each mode corresponds to a different harmonic frequency, which is a multiple of the fundamental frequency $\frac{\pi v}{L}$.

0.2 Fourier Transforms

While Fourier Series deal with periodic functions, the Fourier Transform is used for non-periodic functions. It transforms a function from the time domain to the frequency domain, allowing us to analyze the frequency components of a signal.

Concept of Fourier Transform

For a function $f(t)$ that is not necessarily periodic, the Fourier Transform $F(\omega)$ is defined as:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t}\, dt.$$

The inverse Fourier Transform, which allows us to recover $f(t)$ from its frequency representation, is given by:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t}\, d\omega.$$

The inverse transform is essential because it demonstrates that a function can be fully reconstructed from its frequency components, meaning that no information is lost in the transformation from the time domain to the frequency domain.

Orthogonality of Complex Exponentials

The complex exponentials $e^{i\omega t}$ used in Fourier Transforms are orthogonal over the entire real line:

$$\int_{-\infty}^{\infty} e^{i(\omega-\omega')t}\, dt = 2\pi\delta(\omega-\omega'),$$

where $\delta(\omega-\omega')$ is the Dirac delta function. This property ensures that different frequency components do not interfere with each other, allowing for a clear decomposition of a signal into its frequency parts.

Treatment of Derivatives

Just like with Fourier Series, Fourier Transforms provide a straightforward way to handle derivatives. The Fourier Transform of the derivative $f'(t)$ of a function $f(t)$ is given by:

$$\mathcal{F}\{f'(t)\} = i\omega F(\omega).$$

This shows that taking the derivative in the time domain corresponds to multiplying by $i\omega$ in the frequency domain. This property is extremely useful in solving differential equations, especially in fields like signal processing and quantum mechanics.

Physics Example: Decay of a Radioactive Substance Consider a radioactive substance with a decay constant λ. The number of undecayed atoms $N(t)$ at time t follows an exponential decay law:

$$N(t) = N_0 e^{-\lambda t} \quad (t \geq 0).$$

To analyze the frequency components of this decay process, we take the Fourier Transform of $N(t)$:

$$F(\omega) = \int_0^\infty N_0 e^{-\lambda t} e^{-i\omega t} \, dt.$$

Solving this integral, we find:

$$F(\omega) = \frac{N_0}{\lambda + i\omega}.$$

Key Insight: The Fourier Transform of the exponential decay reveals a Lorentzian distribution in the frequency domain, centered at zero with a width determined by the decay constant λ. This demonstrates how time-domain behavior (exponential decay) is represented in the frequency domain.

Relation Between Space and Momentum, Time and Frequency

In physics, the Fourier Transform reveals a deep relationship between different conjugate variables:

- Time and Frequency: The Fourier Transform decomposes a time-domain signal into its constituent frequencies. This is fundamental in signal processing, where understanding the frequency content of a signal is crucial.

- Space and Momentum: In quantum mechanics, the Fourier Transform relates a particle's wave function in position space $\psi(x)$ to its momentum space representation $\phi(p)$. The position and momentum are Fourier conjugates:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^\infty \psi(x) e^{-ipx/\hbar} \, dx, \quad \psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^\infty \phi(p) e^{ipx/\hbar} \, dp.$$

This duality underpins Heisenberg's uncertainty principle, which states that the more precisely one knows the position, the less precisely one can know the momentum, and vice versa. The Fourier Transform framework provides a natural explanation for this principle: position and momentum are conjugate variables, meaning their respective wavefunctions are related through the Fourier Transform.

Treatment of Derivatives in Space and Momentum Representations:

In quantum mechanics, the treatment of derivatives using Fourier Transforms plays a crucial role in understanding physical phenomena. For example, the momentum operator in the position representation is given by:

$$\hat{p} = -i\hbar \frac{d}{dx}.$$

Applying the Fourier Transform to the derivative in position space transforms this into a multiplication by p in momentum space. This is because:

$$\mathcal{F}\left\{-i\hbar \frac{d}{dx}\psi(x)\right\} = p\phi(p),$$

which is straightforward since taking a derivative in the position domain corresponds to multiplication by $ip/\hbar$ in the momentum domain. This shows that differentiating with respect to position in the physical space translates to a simple multiplication operation in the momentum space.

Relation to Physics:

- Time and Frequency Domain: In signal processing, signals are often represented in the time domain, but their analysis can be more intuitive and revealing in the frequency domain. The Fourier Transform provides a means to switch between these domains seamlessly. For instance, an electrical signal's spectrum can reveal important information about the frequencies that are present, which is crucial for tasks such as filtering or compression.

- Space and Momentum Domain: In quantum mechanics, switching between the space domain (position) and the momentum domain is vital for solving problems involving wave functions. The Fourier Transform facilitates this by allowing the description of a particle's state in either domain, depending on which provides a clearer insight into the physical situation. The relationship between the space and momentum representations of the wave function also provides the mathematical basis for the Heisenberg uncertainty principle, highlighting the intrinsic limitations of simultaneously measuring position and momentum.

Summary: Fourier Analysis is a foundational tool in both classical and quantum physics. It enables the decomposition of complex signals and waveforms into their fundamental components, whether in the context of time and frequency or space and momentum. By understanding the properties of Fourier Series and Fourier Transforms—such as orthogonality, convergence, and the treatment of derivatives—physicists can analyze and interpret a wide range of phenomena, from the vibrations of a string to the behavior of quantum particles.

The discussion of derivatives within Fourier Analysis illustrates how differentiation in the time or spatial domain corresponds to operations in the frequency or momentum domain. This property simplifies solving differential equations and analyzing physical systems, demonstrating the power and versatility of Fourier methods in physics.

Common Fourier Transforms

Below are some common Fourier transforms, along with detailed working for the more complicated examples.

1. Gaussian Function For the Gaussian function $f(t) = e^{-at^2}$, its Fourier Transform is given by:

$$F(\omega) = \int_{-\infty}^{\infty} e^{-at^2} e^{-i\omega t}\, dt.$$

To solve this, we complete the square in the exponent:

$$-at^2 - i\omega t = -a\left(t^2 + \frac{i\omega}{a}t\right).$$

Now, we complete the square:

$$t^2 + \frac{i\omega}{a}t = \left(t + \frac{i\omega}{2a}\right)^2 - \frac{\omega^2}{4a^2}.$$

Substituting back, the integral becomes:

$$F(\omega) = e^{-\frac{\omega^2}{4a}} \int_{-\infty}^{\infty} e^{-a\left(t+\frac{i\omega}{2a}\right)^2}\, dt.$$

The remaining Gaussian integral is standard, and we get:

$$F(\omega) = \sqrt{\frac{\pi}{a}} e^{-\frac{\omega^2}{4a}}.$$

2. Rectangular Function For the rectangular function $f(t)$, defined as:

$$f(t) = \begin{cases} 1 & |t| \leq T/2 \\ 0 & |t| > T/2, \end{cases}$$

the Fourier Transform is:

$$F(\omega) = \int_{-T/2}^{T/2} e^{-i\omega t}\, dt.$$

This is straightforward to integrate. We get:

$$F(\omega) = \left[\frac{e^{-i\omega t}}{-i\omega}\right]_{-T/2}^{T/2} = \frac{e^{-i\omega T/2} - e^{i\omega T/2}}{-i\omega}.$$

Using the identity $e^{ix} - e^{-ix} = 2i\sin(x)$, this simplifies to:

$$F(\omega) = \frac{2\sin(\omega T/2)}{\omega}.$$

Thus, the Fourier Transform of the rectangular function is:

$$F(\omega) = \frac{\sin(\omega T/2)}{\omega/2},$$

which is a sinc function.

3. Exponential Decay For the exponential decay function $f(t) = e^{-\lambda t}$ with $t \geq 0$, the Fourier Transform is:

$$F(\omega) = \int_0^\infty e^{-\lambda t} e^{-i\omega t}\, dt.$$

Factor out the exponential term:

$$F(\omega) = \int_0^\infty e^{-(\lambda + i\omega)t}\, dt.$$

This is a standard exponential integral:

$$F(\omega) = \left[\frac{e^{-(\lambda+i\omega)t}}{-(\lambda + i\omega)} \right]_0^\infty.$$

At the upper limit, the exponential vanishes, and at the lower limit it equals 1, giving:

$$F(\omega) = \frac{1}{\lambda + i\omega}.$$

4. Delta Function The Dirac delta function $\delta(t)$, which is a mathematical idealization representing an impulse at $t = 0$, has the Fourier Transform:

$$F(\omega) = \int_{-\infty}^\infty \delta(t) e^{-i\omega t}\, dt.$$

Since the delta function picks out the value of the integrand at $t = 0$, we get:

$$F(\omega) = e^0 = 1.$$

So the Fourier Transform of the Dirac delta function is:

$$F(\omega) = 1.$$

5. Cosine Function For $f(t) = \cos(\omega_0 t)$, the Fourier Transform is:

$$F(\omega) = \int_{-\infty}^\infty \cos(\omega_0 t) e^{-i\omega t}\, dt.$$

Using Euler's formula $\cos(\omega_0 t) = \frac{e^{i\omega_0 t} + e^{-i\omega_0 t}}{2}$, we split the integral into two parts:

$$F(\omega) = \frac{1}{2} \left(\int_{-\infty}^\infty e^{i(\omega_0 - \omega)t}\, dt + \int_{-\infty}^\infty e^{-i(\omega_0 + \omega)t}\, dt \right).$$

Both of these integrals yield Dirac delta functions:

$$F(\omega) = \frac{1}{2} \left(2\pi\delta(\omega - \omega_0) + 2\pi\delta(\omega + \omega_0) \right).$$

Thus, the Fourier Transform of the cosine function is:

$$F(\omega) = \pi \left(\delta(\omega - \omega_0) + \delta(\omega + \omega_0) \right).$$

This result shows that the Fourier Transform of a cosine function is concentrated at two frequencies, $\omega = \pm\omega_0$, corresponding to the positive and negative frequency components of the cosine wave.

6. Lorentzian Function For the Lorentzian function, also known as the Cauchy distribution, defined as:

$$f(t) = \frac{\gamma}{\pi(t^2 + \gamma^2)},$$

the Fourier Transform is given by:

$$F(\omega) = \int_{-\infty}^{\infty} \frac{\gamma}{\pi(t^2 + \gamma^2)} e^{-i\omega t}\, dt.$$

This is a standard integral, and it can be computed using the residue theorem or recognizing the Fourier Transform of a Lorentzian as a known result. The answer is:

$$F(\omega) = e^{-\gamma|\omega|}.$$

The Lorentzian is important in physics because it describes resonance behavior in systems such as damped harmonic oscillators or signal response functions.